

MUHAMMAD OMER

Data Science Undergraduate | Machine Learning Engineer | Data Analyst

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PROFESSIONAL SUMMARY

BS Data Science student with hands-on experience building end-to-end projects across SQL analytics, computer vision, and machine learning. Conducted advanced SQL analysis on a 100K-order e-commerce dataset (window functions, cohort retention, RFM segmentation) and an HR attrition study surfacing actionable workforce insights. Also built a real-time license plate recognition pipeline with 15–20x faster CPU inference, and a logistics platform combining demand forecasting with linear programming optimization ($R^2 = 0.94$). Seeking a data analyst or data science role where rigorous SQL, Python, and ML skills can drive real business decisions.

EXPERIENCE

Data Analytics Virtual Program, DecodeLabs

May 2026 – Jun 2026

Self-paced virtual job-simulation program

- Cleaned and analyzed real-world datasets, applying EDA and visualization techniques to surface trends across multiple data sources.
- Built visual reports (charts, summary tables) and a final analytics report following a documented, repeatable workflow from raw data to recommendation.

Data Science Virtual Program, Ai Data House

Jul 2025 – Aug 2025

Self-paced virtual job-simulation program

- Applied core data science workflows — data cleaning, exploratory analysis, and model evaluation — to real datasets as part of a guided industry training program.
- Practiced foundational machine learning techniques (regression, classification) and built habits in version control and reproducible analysis.

PROJECTS

Olist E-Commerce SQL Analysis — Data Analyst

2025

MySQL · Advanced SQL · Window Functions · CTEs · RFM Segmentation

- Designed and built a relational MySQL database from scratch across 8 tables (~100K orders), handling messy real-world CSV data including NULL date fields, malformed rows, and missing numeric values.
- Wrote 13 analysis queries answering real business questions: monthly revenue trends, cohort retention, RFM customer segmentation, and late-delivery performance — using CTEs, LAG(), RANK(), NTILE(), and SUM(CASE WHEN).
- Found repeat customers make up only ~3% of the customer base, and November 2017 revenue spiked ~\$988K consistent with Black Friday seasonality — findings documented in a public GitHub repo.

HR Employee Attrition SQL Analysis — Data Analyst

2025

MySQL · Conditional Aggregation · Subqueries · CTEs · Window Functions

- Analyzed IBM's 1,470-employee HR dataset across 10 SQL queries to identify the strongest drivers of employee attrition.
- Discovered Sales Representatives had a 39.76% attrition rate, frequent travelers left at nearly 3x the rate of non-travelers (24.91% vs 8%), and employees in their 20s left at double the rate of those in their 30s.
- Used SUM(CASE WHEN) conditional aggregation, CTE-based subqueries, and RANK() OVER (PARTITION BY) for within-department salary ranking.

Fleet Intelligence Platform — ML Engineer / Operations Research

2024

Python · Scikit-learn · SciPy (Linear Programming) · Streamlit

- Built a logistics control-tower platform forecasting city-zone delivery demand 36 hours ahead with a Random Forest model ($R^2 = 0.94$) using spatial-temporal features like order lag and rolling-mean volume.
- Designed a prescriptive optimization layer with SciPy linear programming to reallocate idle fleet from surplus to deficit zones at minimum cost.
- Translated model and optimizer output into business KPIs — protected revenue and SLA delay reduction — deployed live via Streamlit.

Real-Time License Plate Recognition — Computer Vision / ML Engineer

2024

Python · YOLOv8 · EasyOCR · OpenCV · Flask · Streamlit

- Built an end-to-end AI pipeline detecting vehicles with YOLOv8, localizing plates via Haar Cascades and contour heuristics, and reading text with EasyOCR.
- Achieved ~0.6 seconds/frame on CPU — 15–20x faster than baseline — through async threads, lazy model loading, and resolution capping.
- Deployed a dark-mode web dashboard (Flask + MJPEG stream) with live webcam mode and CLI support, hosted live on Streamlit.

TECHNICAL SKILLS

Languages: Python, SQL, C++

ML / Data: Scikit-learn, TensorFlow, PyTorch, Pandas, Matplotlib, Seaborn, Power BI, Tableau

Tools & Platforms: Flask, Git, Jupyter, VS Code, MongoDB, MS SQL, MySQL

EDUCATION

BS Data Science, PAF-IAST, Haripur

Expected 2028

FSc Computer Science, Jinnah Jam-e School & College

2024

CERTIFICATIONS

Certified Data Scientist — PM Accelerator · Data Analytics Job Simulation — Deloitte · Power BI for Beginners — Simplilearn · SQL (Basic) — HackerRank